Colon Adenocarcinoma (COAD) Exploratory Data Analysis

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Initial Setup

Load packages

```
suppressPackageStartupMessages({
  # BiocManager
  library(GenomicSuperSignature)
  library(curatedTCGAData)
  library(MultiAssayExperiment)
  library(TCGAutils)
  library(ComplexHeatmap)

  # CRAN
  library(tidyverse) # includes dplyr, ggplot2, magrittr, tidyr
  library(magick)
  library(wordcloud)
  library(ztable)
  library(metafolio)
})
```

Load RAVmodel

```
RAVmodel <- getModel('C2', load=TRUE)
```

```
## [1] "downloading"
```

Select COAD RNA metadata

```
coad <- curatedTCGAData(diseaseCode = 'COAD',
                        assays = 'RNA*',
                        version = '2.0.1',
                        dry.run = FALSE)

coad_rna <- getWithColData(coad,
```

```
                              'COAD_RNASeq2Gene-20160128')

assay(coad_rna) <- log2(assay(coad_rna) + 1)

coad_meta <- colData(coad_rna)
head(colData(coad_rna)[,1:4])
```

```
## DataFrame with 6 rows and 4 columns
##                 patientID years_to_birth vital_status.x days_to_death.x
##               <character>      <integer>      <integer>       <integer>
## TCGA-3L-AA1B TCGA-3L-AA1B             61              0              NA
## TCGA-4N-A93T TCGA-4N-A93T             67              0              NA
## TCGA-4T-AA8H TCGA-4T-AA8H             42              0              NA
## TCGA-5M-AAT4 TCGA-5M-AAT4             74              1              49
## TCGA-5M-AAT6 TCGA-5M-AAT6             40              1             290
## TCGA-5M-AATE TCGA-5M-AATE             76              0              NA
```

```
sampleTables(coad)
```

```
## $`COAD_RNASeq2Gene-20160128`
##
##  01  02  06  11
## 283   1   1  41
##
## $`COAD_RNASeq2GeneNorm_illuminaga-20160128`
##
##  01
## 191
##
## $`COAD_RNASeq2GeneNorm_illuminahiseq-20160128`
##
##  01  02  06  11
## 283   1   1  41
##
## $`COAD_RNASeqGene-20160128`
##
## 01
## 10
```

```
head(getSubtypeMap(coad))
```

```
##        COAD_annotations        COAD_subtype
## 1            Patient_ID           patientID
## 2                   msi          MSI_status
## 3  methylation_subtypes methylation_subtype
## 4         mrna_subtypes  expression_subtype
## 5 histological_subtypes   histological_type
```

```
getClinicalNames("COAD")
```

```
##  [1] "years_to_birth"
```

```
##  [2] "vital_status"
##  [3] "days_to_death"
##  [4] "days_to_last_followup"
##  [5] "tumor_tissue_site"
##  [6] "pathologic_stage"
##  [7] "pathology_T_stage"
##  [8] "pathology_N_stage"
##  [9] "pathology_M_stage"
## [10] "gender"
## [11] "date_of_initial_pathologic_diagnosis"
## [12] "days_to_last_known_alive"
## [13] "radiation_therapy"
## [14] "histological_type"
## [15] "residual_tumor"
## [16] "number_of_lymph_nodes"
## [17] "race"
## [18] "ethnicity"
```

```
coad2 <- TCGAsplitAssays(coad, c("01", "11"))

coad_rna_cancer <- getWithColData(coad2,
                            '01_COAD_RNASeq2Gene-20160128')
assay(coad_rna_cancer) <- log2(assay(coad_rna_cancer) + 1)
coad_cancer <- colData(coad_rna_cancer)

coad_rna_normal <- getWithColData(coad2,
                            '11_COAD_RNASeq2Gene-20160128')
assay(coad_rna_normal) <- log2(assay(coad_rna_normal) + 1)
coad_normal <- colData(coad_rna_normal)

table(coad_cancer$patient.microsatellite_instability)
```

```
##
##  no yes
##  75  10
```

```
table(coad_normal$patient.microsatellite_instability)
```

```
##
##  no yes
##   5   1
```

```
coad_normal@rownames %in% coad_cancer@rownames
```

```
##  [1] FALSE  TRUE FALSE FALSE FALSE  TRUE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE
## [13]  TRUE  TRUE  TRUE  TRUE  TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [25] FALSE FALSE FALSE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE  TRUE
## [37]  TRUE  TRUE  TRUE  TRUE  TRUE
```

```
normal_tb <- coad_normal %>%
  as.data.frame() %>%
```

```
  select(patientID, patient.microsatellite_instability)

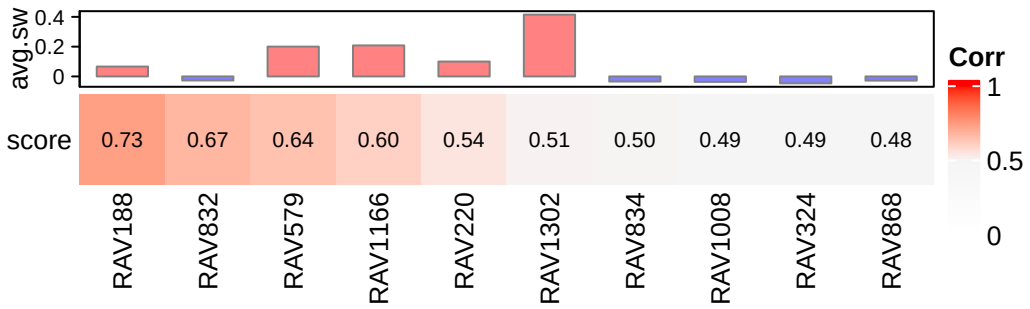
cancer_tb <- coad_cancer %>%
  as.data.frame() %>%
  select(patientID, patient.microsatellite_instability)

combined_tb <- inner_join(normal_tb, cancer_tb, by="patientID")
#Remove these 4 rows: TCGA-A6-5662, TCGA-A6-5665, TCGA-A6-5667, TCGA-F4-6704
mss_patient_id <- c("TCGA-A6-5662", "TCGA-A6-5665", "TCGA-A6-5667", "TCGA-F4-6704")

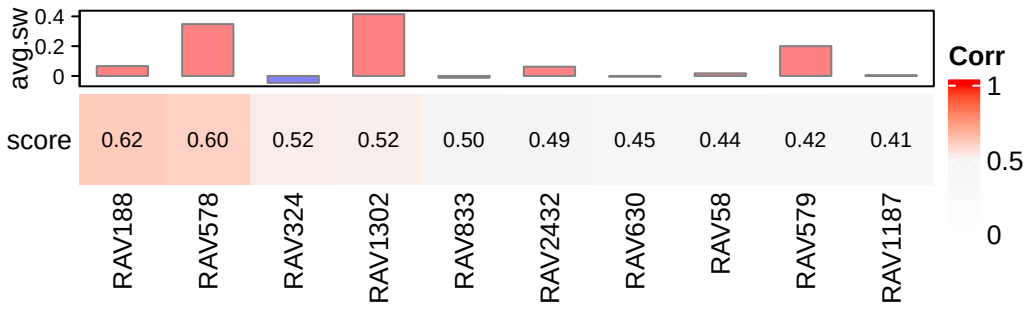
coad_normal <- coad_normal[-which(coad_normal@listData[["patientID"]] %in% mss_patient_id), ]
coad_cancer <- coad_cancer[-which(coad_cancer@listData[["patientID"]] %in% mss_patient_id), ]
```

heatmapTable: COAD

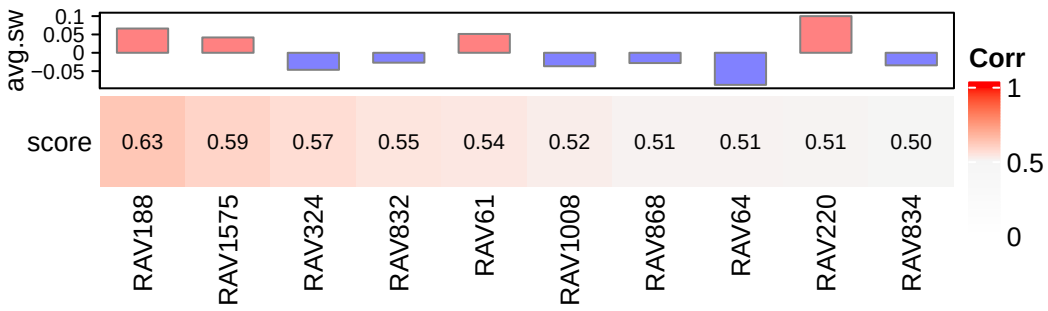
```
validate_coad_rna <- validate(coad_rna, RAVmodel)
validate_coad_normal <- validate(coad_rna_normal, RAVmodel)
validate_coad_cancer <- validate(coad_rna_cancer, RAVmodel)
heatmapTable(validate_coad_rna, RAVmodel, num.out = 10)
```




```
heatmapTable(validate_coad_normal, RAVmodel, num.out = 10)
```




```
heatmapTable(validate_coad_cancer, RAVmodel, num.out=10)
```

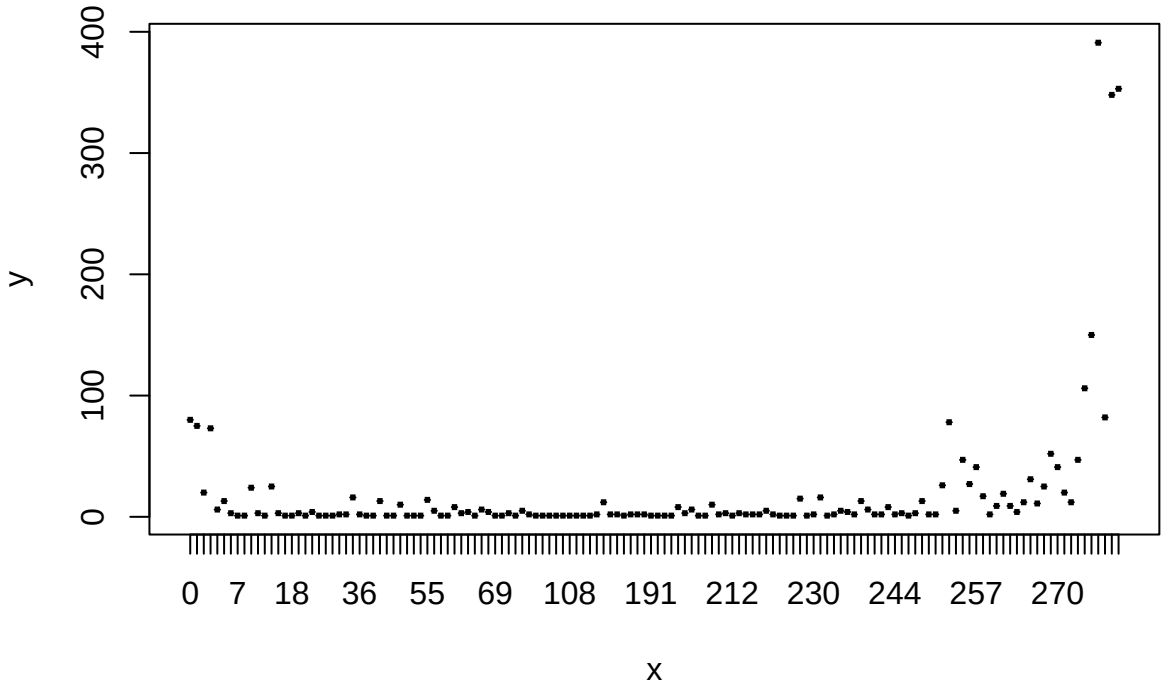


Subset ## Filter attributes

```
sparsity_summary <- table(colSums(is.na(coad_cancer)))
sparsity_summary
```

```
##
##   0   1   2   3   4   5   6   7   8   9  10  11  13  14  16  18  19  20  23  25
##  80  75  20  73   6  13   3   1   1  24   3   1  25   3   1   1   3   1   4   1
##  30  32  33  34  35  36  37  38  40  41  43  44  48  49  54  55  57  61  62  63
##   1   1   2   2  16   2   1   1  13   1   1  10   1   1   1  14   5   1   1   8
##  64  65  66  67  68  69  72  79  81  94  95  96  97  99 104 106 108 138 147 148
##   3   4   1   6   4   1   1   3   1   5   2   1   1   1   1   1   1   1   1   1
## 151 175 176 177 179 183 188 190 191 195 198 201 203 204 206 207 208 209 210 211
##   2  12   2   2   1   2   2   2   1   1   1   1   8   3   6   1   1  10   2   3
## 212 213 216 219 220 221 224 225 226 227 228 229 230 231 233 234 235 236 237 238
##   1   3   2   2   2   5   2   1   1   1  15   1   2  16   1   2   5   4   2  13
## 239 240 242 243 244 245 246 247 248 249 251 252 253 254 255 256 257 258 260 261
##   6   2   2   8   2   3   1   3  13   2   2  26  78   5  47  27  41  17   2   9
## 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279
##  19   9   4  12  31  11  25  52  41  20  12  47 106 150 391  82 348 353
```

Sparsity Plot




```
# Select columns with >10% completeness
keep_attribute_ind <- which(colSums(!is.na(coad_cancer)) > round(nrow(coad_cancer)/10))
meta_sub1 <- coad_cancer[keep_attribute_ind]
meta_sub1 <- subset(coad_cancer, select= -patientID)
```

```
# Randomly select for 140 rows
set.seed(1)
random_sample_ind <- sample(1:nrow(meta_sub1), 140)
meta_sub2 <- meta_sub1[random_sample_ind,]
```

```
# Check for data types in listData
unique(sapply(coad_cancer@listData, type))
```

```
## [1] "character" "integer"   "double"
```

```
charcTb <- meta_sub2[, sapply(meta_sub2, class) == 'character']
numTb <- meta_sub2[, sapply(meta_sub2, class) %in% c('numeric', 'integer')]

# Sort numeric variables that have <=5 unique values as character variables
# addToFactors <- c()
# for (i in 1:length(numTb)) {
#   if (length(table(numTb[i])) <= 3) {
```

```
#     addToFactors <- c(addToFactors, i)
#   }
# }

# charcTb <- c(charcTb, numTb[addToFactors])
# numTb <- numTb[-addToFactors]
```

```
# Calculate validation scores
sampleScore <- calculateScore(coad_rna_cancer, RAVmodel)
```

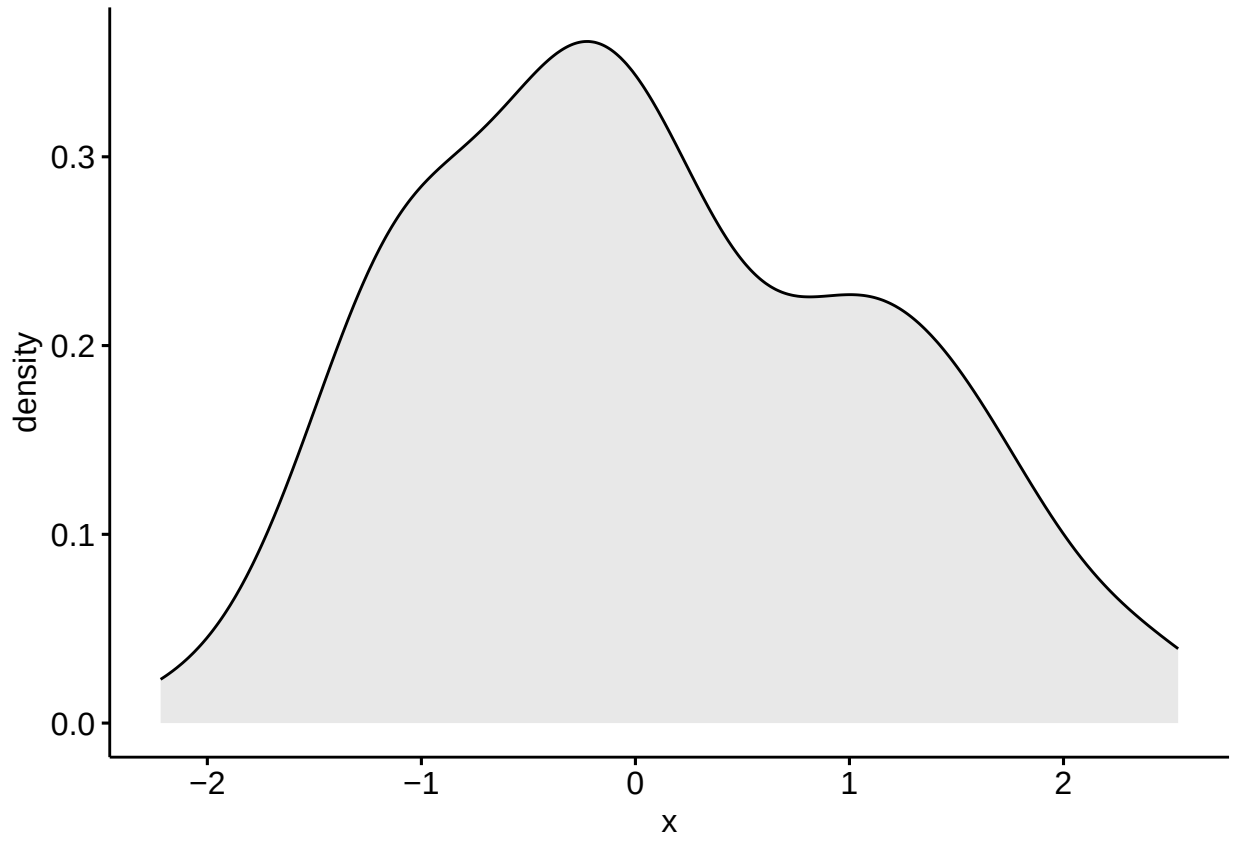
```
validated_ind <- validatedSignatures(validate_coad_cancer, RAVmodel, num.out = 15, scoreCutoff = 0.45, i
```

```
## RAV61 can be filtered based on GSEA_C2
```

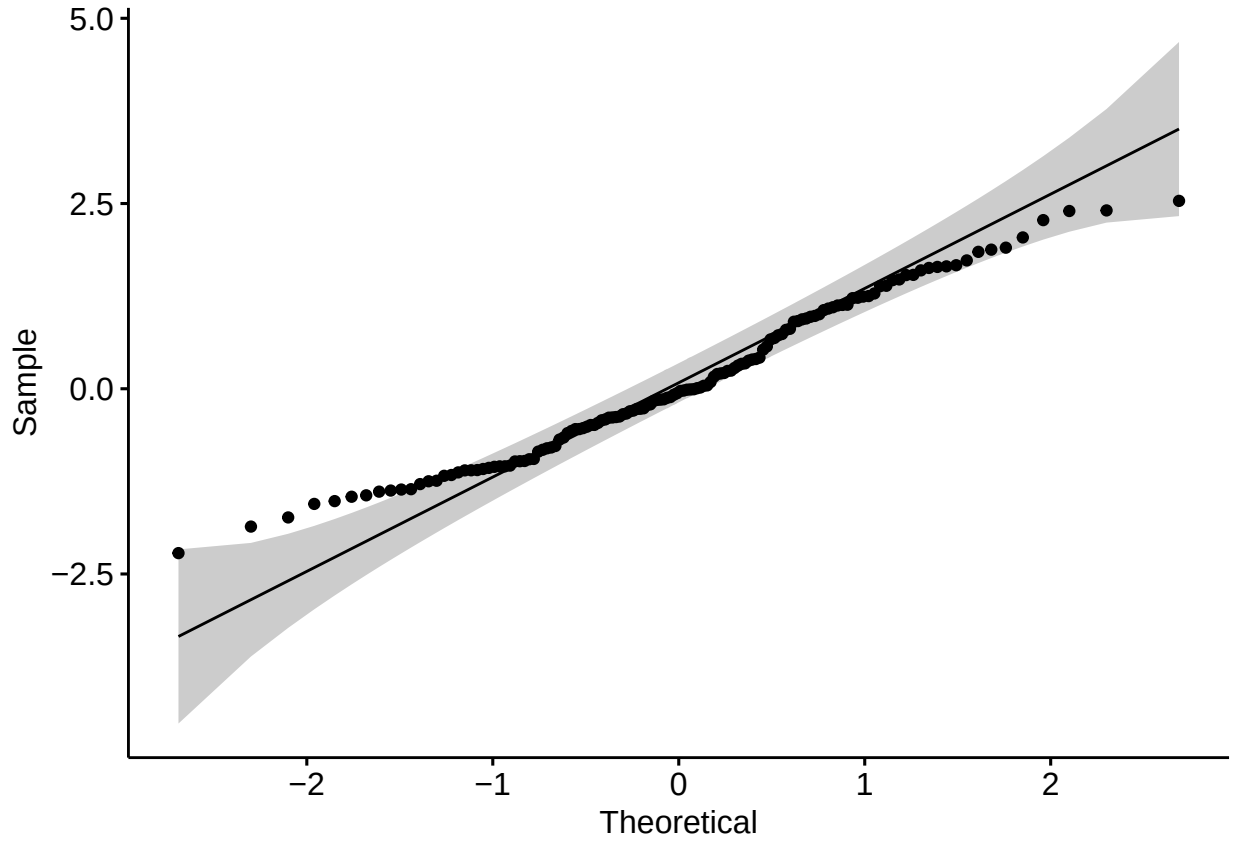
```
## RAV438 can be filtered based on GSEA_C2
```

```
# Subset sampleScore to join with MCPcounter
sampleScore_sub <- sampleScore[random_sample_ind, validated_ind] %>% as.data.frame()
```

```
library("ggpubr")
# Density plot
ggdensity(sampleScore_sub$RAV834, fill = "lightgray")
```



```r
# QQ plot
ggqqplot(sampleScore_sub$RAV834)
```



Calculate R-Squared Value for Numeric Variables

```r
# R squared value function
calculateRsq <- function (x, y) stats::cor(x, y, use = 'na.or.complete') ^ 2
```

```r
# Calculate r-squared for numeric attributes
rsq_numAttr <- as.data.frame(matrix(nrow = ncol(numTb),
                                    ncol = ncol(sampleScore_sub)))

colnames(rsq_numAttr) <- colnames(sampleScore_sub)
rownames(rsq_numAttr) <- colnames(numTb)

for (i in seq_len(ncol(numTb))) {
  for (j in seq_len(ncol(sampleScore_sub))) {
    rsq <- calculateRsq(numTb[, i], sampleScore_sub[, j])
    rsq_numAttr[i, j] <- rsq
  }
}

rsq_numAttr <- na.omit(rsq_numAttr)
```

```
# Remove batch effect variables
batch_num_ind <- grep('analyte|portion|procurement|aliquot|uuid|barcode',
                      rownames(rsq_numAttr))
rsq_numAttr_no_batch <- rsq_numAttr[-batch_num_ind,]
```

```
max_rav <- apply(rsq_numAttr_no_batch, 1, max)
max_attr <- which(max_rav > 0.5)

target_rsq <- rsq_numAttr_no_batch[max_attr,]
```

heatmapTable

```
options(ztable.type='html')
z = ztable(target_rsq)
z %>% makeHeatmap(palette="Blues")
```

days_to_last_known_alive.x

0.04

0.02

0.23

0.43

0.05

0.37

0.01

0.00

0.02

0.59

0.14

0.03

0.19

0.13

0.01

patient.days_to_last_known_alive

0.04

0.02

0.23

0.43

0.05

0.37

0.01

0.00

0.02

0.59

0.14

0.03

0.19

0.13

0.01

patient.drugs.drug.10.day_of_form_completion

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.10.days_to_drug_therapy_end

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.10.days_to_drug_therapy_start

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.10.month_of_form_completion

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.10.number_cycles

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.10.regimen_number

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.10.total_dose

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.5.number_cycles

0.21

0.53

0.02

0.06

0.55

0.20

0.03

0.36

0.24

0.00

0.00

0.40

0.22

0.04

0.02

patient.drugs.drug.5.total_dose

0.45

0.40

0.11

0.02

0.32

0.29

0.13

0.35

0.54

0.12

0.23

0.00

0.23

0.11

0.17

patient.drugs.drug.6.days_to_drug_therapy_end

0.00

0.00

0.01

0.55

0.01

0.01

0.04

0.00

0.00

0.22

0.01

0.06

0.04

0.17

0.09

patient.drugs.drug.6.number_cycles

0.53

0.69

0.08

0.02

0.78

0.52

0.00

0.55

0.72

0.36

0.05

0.37

0.74

0.00

0.05

patient.drugs.drug.6.regimen_number

0.17

0.27

0.00

0.02

0.50

0.22

0.12

0.08

0.33

0.49

0.00

0.54

0.59

0.17

0.05

patient.drugs.drug.6.total_dose

0.42

0.37

0.37

0.33

0.75

0.30

0.69

0.16

0.08

0.08

0.42

0.74

0.31

0.99

0.34

patient.drugs.drug.7.days_to_drug_therapy_end

0.09

0.15

0.05

0.79

0.06

0.16

0.11

0.16

0.15

0.19

0.03

0.00

0.01

0.43

0.08

patient.drugs.drug.7.days_to_drug_therapy_start

0.09

0.18

0.07

0.80

0.06

0.17

0.16

0.18

0.18

0.22

0.06

0.02

0.01

0.49

0.14

patient.drugs.drug.7.number_cycles

0.05

0.00

0.18

0.11

0.05

0.00

0.72

0.00

0.00

0.42

0.34

0.45

0.15

0.78

0.81

patient.drugs.drug.7.regimen_number

0.22

0.31

0.05

0.97

0.26

0.33

0.02

0.23

0.24

0.00

0.07

0.13

0.15

0.26

0.00

patient.drugs.drug.7.total_dose

0.15

0.30

0.00

0.90

0.34

0.32

0.03

0.21

0.24

0.03

0.00

0.34

0.26

0.06

0.10

patient.drugs.drug.8.days_to_drug_therapy_end

0.01

0.00

0.00

0.73

0.09

0.00

0.15

0.02

0.01

0.83

0.00

0.53

0.21

0.40

0.27

patient.drugs.drug.8.days_to_drug_therapy_start

0.07

0.06

0.02

0.50

0.27

0.05

0.08

0.01

0.02

0.96

0.01

0.59

0.44

0.24

0.22

patient.drugs.drug.8.month_of_form_completion

0.09

0.20

0.12

0.96

0.05

0.22

0.23

0.34

0.31

0.34

0.03

0.22

0.00

0.59

0.22

patient.drugs.drug.8.number_cycles

0.27

0.45

0.13

0.64

0.88

0.44

0.04

0.48

0.55

1.00

0.03

0.65

0.99

0.11

0.20

patient.drugs.drug.8.regimen_number

0.93

0.99

0.81

0.03

0.85

0.99

0.27

1.00

1.00

0.48

0.64

0.03

0.65

0.16

0.08

patient.drugs.drug.8.total_dose

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.8.year_of_form_completion

0.17

0.38

0.16

0.77

0.27

0.40

0.11

0.56

0.55

0.06

0.02

0.02

0.16

0.42

0.06

patient.drugs.drug.9.day_of_form_completion

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.9.days_to_drug_therapy_end

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

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1.00

1.00

patient.drugs.drug.9.days_to_drug_therapy_start

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

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1.00

patient.drugs.drug.9.month_of_form_completion

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.9.number_cycles

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.drugs.drug.9.regimen_number

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.follow_ups.follow_up.2.days_to_death

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event.days_to_new_tumor_event_additional_surgery_p

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event.days_to_new_tumor_event_after_initial_treatmer

0.07

0.03

0.21

0.34

0.00

0.09

0.34

0.10

0.14

0.02

0.20

0.52

0.00

0.30

0.11

patient.follow_ups.follow_up.3.day_of_form_completion

0.33

0.32

0.19

0.22

0.33

0.21

0.41

0.32

0.43

0.02

0.26

0.21

0.21

0.32

0.53

patient.follow_ups.follow_up.3.days_to_last_followup

0.03

0.02

0.10

0.24

0.00

0.03
0.22
0.05
0.03
0.31
0.09
0.59
0.01
0.23
0.18
patient.follow_ups.follow_up.3.month_of_form_completion
0.00
0.00
0.07
0.48
0.04
0.01
0.17
0.00
0.01
0.12
0.07
0.59
0.02
0.11
0.07
patient.follow_ups.follow_up.3.new_tumor_events.new_tumor_event.days_to_new_tumor_event_after_initial_treatmer
1.00
1.00
1.00
1.00
1.00
1.00
1.00
1.00
1.00

1.00

1.00

1.00

1.00

1.00

1.00

patient.follow_ups.follow_up.3.year_of_form_completion

0.89

0.86

0.79

0.21

0.82

0.82

0.89

0.86

0.94

0.06

0.85

0.52

0.76

0.82

0.88

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.days_to_new_tumor_event_additional_surgery_pro

0.55

0.71

0.38

0.27

0.58

0.55

0.26

0.71

0.74

0.26

0.24

0.04

0.47

0.74

0.54

patient.new_tumor_events.new_tumor_event.days_to_new_tumor_event_additional_surgery_procedure

0.92

0.93

0.95

1.00

0.88

0.81

0.97

0.95

0.97

0.74

0.79

0.98

0.76

1.00

0.95

patient.new_tumor_events.new_tumor_event.days_to_new_tumor_event_after_initial_treatment

0.72

0.92

0.78

0.01

0.95

0.96

0.81

0.90

0.95

0.55

0.57

0.61

0.83

0.83

0.73

patient.radiations.radiation.course_number

0.17

0.05
0.38
0.16
0.06
0.04
0.59
0.09
0.03
0.65
0.38
0.99
0.06
0.42
0.73
patient.radiations.radiation.radiation_dosage
0.26
0.12
0.34
0.06
0.00
0.07
0.50
0.30
0.18
0.05
0.38
0.73
0.04
0.30
0.40
patient.biospecimen_cqcf.tumor_samples.tumor_sample.2.tumor_necrosis_percent
0.90
0.94
0.44
0.00
0.83

0.22

0.32

0.97

0.20

0.98

0.89

0.98

0.00

0.28

0.01

patient.biospecimen_cqcf.tumor_samples.tumor_sample.2.tumor_nuclei_percent

0.40

0.32

0.22

0.92

0.50

0.98

0.32

0.25

0.99

0.23

0.00

0.05

0.84

0.96

0.94

patient.biospecimen_cqcf.tumor_samples.tumor_sample.2.tumor_weight

0.52

0.60

0.84

0.13

0.42

0.00

0.75

0.68

0.00

0.70

0.99

0.91

0.22

0.01

0.10

patient.biospecimen_cqcf.tumor_samples.tumor_sample.2.vial_number

0.64

0.56

0.06

0.74

0.73

0.99

0.13

0.48

0.98

0.46

0.07

0.20

0.63

1.00

0.78

patient.samples.sample.2.initial_weight

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

SNP6

0.67

0.50

0.20

0.19

0.20

0.30

0.07

0.64

0.41

0.09

0.17

0.01

0.27

0.37

0.10

methylation

0.34

0.65

0.12

0.03

0.61

0.69

0.01

0.50

0.49

0.07

0.01

0.05

0.76

0.36

0.00

silent_mutrate

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

nonsilent_mutrate

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

1.00

number_of_lymphnodes_positive_by_he

0.88

0.56

0.21

0.17

0.19

0.35

0.05

0.73

0.62

0.11

0.16

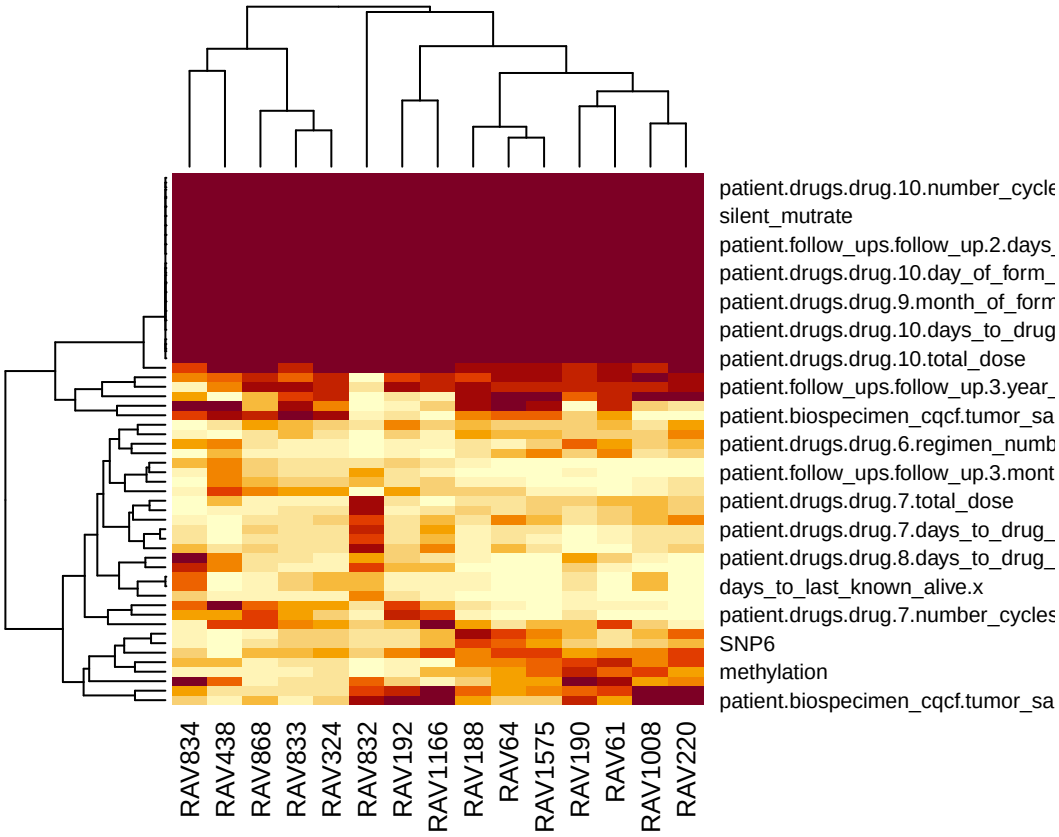
0.00

0.29

0.38

0.09

```
heatmap(as.matrix(target_rsq), scale="column")
```



Separate out Factor Variables by number of levels (1, 2, 3+) not including NA

```r
# Convert to factor data type
factorTb <- meta_sub2[, sapply(meta_sub1, class) == 'character']

factorTb[sapply(factorTb, is.character)] <- lapply(factorTb[sapply(factorTb, is.character)], factor, ex

#any(is.na(levels(factorTb[,2])))
#!any(is.na(levels(factorTb[,6])))

single_fac_ind <- c()
binary_fac_ind <- c()
multi_fac_ind <- c()

# Testing factor grouping
for (i in 1:length(factorTb)) {
  if (nlevels(factorTb[,i]) == 1 |
      (nlevels(factorTb[,i]) == 2 &
       any(is.na(levels(factorTb[,i]))))
      ) {
    single_fac_ind <- c(single_fac_ind, i)
  } else if (#nlevels(factorTb[,i]) == 3 &
             #any(is.na(levels(factorTb[,i]))) |

             (nlevels(factorTb[,i]) == 2 &
              !any(is.na(levels(factorTb[,i]))))
           ) {
    binary_fac_ind <- c(binary_fac_ind, i)
  } else {
    multi_fac_ind <- c(multi_fac_ind, i)
  }
}

new_factorTb <- factorTb[,multi_fac_ind]
binary_factor <- factorTb[,binary_fac_ind]
#binary_factor <- binary_factor[-c(7,8)]
single_factor <- factorTb[,single_fac_ind]
```

Calculate F-statistic (ANOVA) for Character Variables

```r
# aov_res <- as.data.frame(matrix(nrow = ncol(new_factorTb),
#                                 ncol = ncol(sampleScore_sub)))
#
# rownames(aov_res) <- colnames(new_factorTb)
# colnames(aov_res) <- colnames(sampleScore_sub)
#
# aov_coad_fvalue <- aov_res
# aov_coad_pvalue <- aov_res
#
# for (i in seq_len(ncol(sampleScore_sub))) {
#   for (j in seq_len(ncol(new_factorTb))) {
#
#     ## ANOVA
```

```
#     aov <- aov(sampleScore_sub[, i] ~ new_factorTb[, j])
#
#     ## F-statistic
#     fval <- summary(aov)[[1]]$`F value`[1]
#     aov_coad_fvalue[j, i] <- fval
#
#     ## p-value
#     pval <- summary(aov)[[1]]$`Pr(>F)`[1]
#     aov_coad_pvalue[j, i] <- pval
#   }
# }
```

```
# Select for p-values < 0.01
# min_rav <- apply(aov_coad_pvalue, 1, min)
# min_attr <- which(min_rav < 0.01)
#
# target_coad_aov_fvalue <- aov_coad_fvalue[min_attr,]
# target_coad_aov_pvalue <- aov_coad_pvalue[min_attr,]
```

```
# batch_char_ind <- grep('analyte|portion|procurement|aliquot|uuid|barcode',
#                        rownames(target_coad_aov_fvalue))
# coad_aov_fvalue <- target_coad_aov_fvalue[-batch_char_ind,]
# coad_aov_pvalue <- target_coad_aov_pvalue[-batch_char_ind,]
```

```
# heatmap(as.matrix(coad_aov_fvalue), main = 'COAD F-Statistics')
```

```
sig_fval <- as.data.frame(matrix(nrow = ncol(new_factorTb),
                                 ncol = ncol(sampleScore_sub)))

rownames(sig_fval) <- colnames(new_factorTb)
colnames(sig_fval) <- colnames(sampleScore_sub)

for (i in seq_len(ncol(sampleScore_sub))) {
  for (j in seq_len(ncol(new_factorTb))) {

    if (!is.null(summary(aov(sampleScore_sub[, i] ~ new_factorTb[, j]))[[1]]$`Pr(>F)`[1])) {

      if (!is.null(summary(aov(sampleScore_sub[, i] ~ new_factorTb[, j]))[[1]]$`F value`[1]) &
          (summary(aov(sampleScore_sub[, i] ~ new_factorTb[, j]))[[1]]$`Pr(>F)`[1] < 0.05)) {
        sig_fval[j, i] <- summary(aov(sampleScore_sub[, i] ~ new_factorTb[, j]))[[1]]$`F value`[1]
      } else {
        next
      }
    } else {
      next
    }
  }
}

fstat_ind <- c()

for (i in seq_len(nrow(sig_fval))) {
```

```
  if (sum(is.na(sig_fval[i, ])) < 220) {
    fstat_ind <- c(fstat_ind, i)
  } else {
    next
  }
}

sig_fval <- sig_fval[fstat_ind,]
batch_char_ind_2 <- grep('analyte|portion|procurement|aliquot|uuid|barcode',
                  rownames(sig_fval))
sig_fval <- sig_fval[-batch_char_ind_2,]
```

```
options(ztable.type='html')
ztable(sig_fval) %>%
makeHeatmap(palette = 'Blues') %>%
print(caption='ANOVA F-stats with p-values < 0.01')
```

ANOVA F-stats with p-values < 0.01

pathologic_stage

pathology_T_stage

pathology_N_stage

pathology_M_stage

radiation_therapy

4.06

6.93

7.04

4.82

7.13

histological_type.x

3.18

4.06

4.37

3.21

residual_tumor.x

race

ethnicity

4.02

4.19

3.90

3.64

4.74

patient.anatomic_neoplasm_subdivision

patient.braf_gene_analysis_performed

3.56

3.82

5.72

4.46

8.33

4.80

5.14

3.96

6.97

8.16

patient.braf_gene_analysis_result

3.39

3.94

5.85

4.97

8.37

4.73

5.16

4.19

7.62

8.28

patient.clinical_cqcf.country

2.15

2.42

2.55

2.48

2.46

2.39

2.91

patient.clinical_cqcf.frozen_specimen_anatomic_site

patient.clinical_cqcf.histological_type

3.20

patient.clinical_cqcf.history_of_prior_malignancy

3.60

3.93

patient.clinical_cqcf.normal_tissue_anatomic_site

patient.colon_polyps_present

patient.drugs.drug.10.drug_name

patient.drugs.drug.10.prescribed_dose

patient.drugs.drug.10.prescribed_dose_units

patient.drugs.drug.10.total_dose_units

patient.drugs.drug.2.drug_name

patient.drugs.drug.2.measure_of_response

patient.drugs.drug.2.prescribed_dose

1.66

patient.drugs.drug.2.prescribed_dose_units

patient.drugs.drug.2.regimen_indication

patient.drugs.drug.2.route_of_administrations.route_of_administration

patient.drugs.drug.2.therapy_ongoing

patient.drugs.drug.2.therapy_types.therapy_type

patient.drugs.drug.2.therapy_types.therapy_type_notes

patient.drugs.drug.2.total_dose_units

patient.drugs.drug.2.tx_on_clinical_trial

patient.drugs.drug.3.drug_name

1.98

patient.drugs.drug.3.measure_of_response

3.04

patient.drugs.drug.3.prescribed_dose

1.64

patient.drugs.drug.3.prescribed_dose_units

patient.drugs.drug.3.regimen_indication

patient.drugs.drug.3.route_of_administrations.route_of_administration

patient.drugs.drug.3.therapy_ongoing

patient.drugs.drug.3.therapy_types.therapy_type

patient.drugs.drug.3.therapy_types.therapy_type_notes

patient.drugs.drug.3.total_dose_units

2.85

2.75

patient.drugs.drug.3.tx_on_clinical_trial

patient.drugs.drug.4.drug_name

patient.drugs.drug.4.measure_of_response

3.15

3.63

patient.drugs.drug.4.prescribed_dose

patient.drugs.drug.4.regimen_indication

patient.drugs.drug.4.therapy_ongoing

patient.drugs.drug.4.therapy_types.therapy_type

patient.drugs.drug.4.therapy_types.therapy_type_notes

2.76

2.54

2.63

patient.drugs.drug.4.tx_on_clinical_trial

patient.drugs.drug.5.drug_name

patient.drugs.drug.5.measure_of_response

3.75

4.06

patient.drugs.drug.5.prescribed_dose

patient.drugs.drug.5.regimen_indication

patient.drugs.drug.5.therapy_ongoing

patient.drugs.drug.5.therapy_types.therapy_type

patient.drugs.drug.5.therapy_types.therapy_type_notes

3.21

4.13

3.48

patient.drugs.drug.5.tx_on_clinical_trial

patient.drugs.drug.6.drug_name

2.42

2.25

2.44

patient.drugs.drug.6.measure_of_response

3.75

4.06

patient.drugs.drug.6.prescribed_dose

patient.drugs.drug.6.regimen_indication

patient.drugs.drug.6.therapy_ongoing

patient.drugs.drug.6.therapy_types.therapy_type

4.87

3.07

4.68

3.68

4.25

4.93

patient.drugs.drug.7.drug_name

patient.drugs.drug.7.prescribed_dose

patient.drugs.drug.7.regimen_indication

3.44

patient.drugs.drug.7.therapy_types.therapy_type

2.83

2.94

2.80

patient.drugs.drug.8.drug_name

patient.drugs.drug.8.prescribed_dose

patient.drugs.drug.8.regimen_indication

patient.drugs.drug.9.drug_name

patient.drugs.drug.drug_name

patient.drugs.drug.measure_of_response

patient.drugs.drug.number_cycles

patient.drugs.drug.prescribed_dose

patient.drugs.drug.prescribed_dose_units

patient.drugs.drug.regimen_indication

patient.drugs.drug.regimen_number

patient.drugs.drug.route_of_administrations.route_of_administration

patient.drugs.drug.therapy_ongoing

patient.drugs.drug.therapy_types.therapy_type

patient.drugs.drug.therapy_types.therapy_type_notes

patient.drugs.drug.total_dose

patient.drugs.drug.total_dose_units

patient.drugs.drug.tx_on_clinical_trial

4.25

5.80

4.87

4.53

4.75

3.84

3.15

4.54

patient.ethnicity

4.02

4.19

3.90

3.64

4.74

patient.follow_ups.follow_up.2.followup_case_report_form_submission_reason

patient.follow_ups.follow_up.2.followup_treatment_success

patient.follow_ups.follow_up.2.lost_follow_up

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event.additional_pharmaceutical_therapy

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event.additional_radiation_therapy

3.69

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event.new_tumor_event_additional_surgery_procedure

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event.site_of_additional_surgery_new_tumor_event_me

patient.follow_ups.follow_up.2.new_tumor_events.new_tumor_event_after_initial_treatment

patient.follow_ups.follow_up.2.person_neoplasm_cancer_status

patient.follow_ups.follow_up.2.postoperative_rx_tx

patient.follow_ups.follow_up.2.radiation_therapy

3.27

patient.follow_ups.follow_up.2.vital_status

patient.follow_ups.follow_up.3.new_tumor_events.new_tumor_event_after_initial_treatment

patient.follow_ups.follow_up.3.person_neoplasm_cancer_status

patient.follow_ups.follow_up.3.vital_status

patient.follow_ups.follow_up.followup_case_report_form_submission_reason

patient.follow_ups.follow_up.followup_treatment_success

patient.follow_ups.follow_up.lost_follow_up

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.additional_pharmaceutical_therapy

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.additional_radiation_therapy

3.12

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.new_neoplasm_event_type

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.new_tumor_event_additional_surgery_procedure

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.residual_disease_post_new_tumor_event_margin_s

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event.site_of_additional_surgery_new_tumor_event_mets

patient.follow_ups.follow_up.new_tumor_events.new_tumor_event_after_initial_treatment

patient.follow_ups.follow_up.person_neoplasm_cancer_status

4.25

3.09

3.67

patient.follow_ups.follow_up.postoperative_rx_tx

4.51

3.29

7.06

3.74

4.31

5.39

7.19

9.05

patient.follow_ups.follow_up.primary_therapy_outcome_success

patient.follow_ups.follow_up.radiation_therapy

4.35

7.56

3.52

7.39

6.19

8.45

patient.follow_ups.follow_up.vital_status

patient.histological_type

3.18

4.06

4.37

3.21

patient.history_of_colon_polyps

patient.icd_10

patient.icd_o_3_histology

5.37

patient.icd_o_3_site

patient.kras_gene_analysis_performed

3.45

3.40

3.07

5.36

3.57

3.82

6.06

5.99

patient.kras_mutation_found

3.51

4.73

3.92

4.13

6.47

4.63

5.03

3.21

7.46

7.11

patient.loss_expression_of_mismatch_repair_proteins_by_ihc

3.12

5.23

4.83

4.77

patient.loss_expression_of_mismatch_repair_proteins_by_ihc_results.loss_expression_of_mismatch_repair_proteins_by

4.21

6.82

9.00

3.88

4.02

patient.loss_expression_of_mismatch_repair_proteins_by_ihc_results.loss_expression_of_mismatch_repair_proteins_by

3.86

6.30

9.91

3.28

4.33

patient.loss_expression_of_mismatch_repair_proteins_by_ihc_results.loss_expression_of_mismatch_repair_proteins_by

5.78

6.24

3.96

7.56

3.53

patient.loss_expression_of_mismatch_repair_proteins_by_ihc_results.loss_expression_of_mismatch_repair_proteins_by

3.72

6.20

9.68

3.29

4.23

patient.lymphatic_invasion

patient.microsatellite_instability

3.76

patient.new_tumor_events.new_tumor_event.additional_pharmaceutical_therapy

patient.new_tumor_events.new_tumor_event.additional_radiation_therapy

patient.new_tumor_events.new_tumor_event.new_neoplasm_event_type

patient.new_tumor_events.new_tumor_event.new_tumor_event_additional_surgery_procedure

3.44

patient.new_tumor_events.new_tumor_event_after_initial_treatment

3.45

patient.non_nodal_tumor_deposits

patient.patient_id

patient.perineural_invasion_present

patient.person_neoplasm_cancer_status

5.51

3.36

7.85

3.20

4.16

4.51

5.99

9.61

patient.postoperative_rx_tx

3.13

patient.primary_lymph_node_presentation_assessment

4.70

patient.primary_therapy_outcome_success

patient.race

patient.radiations.radiation.anatomic_treatment_site

2.98

patient.radiations.radiation.measure_of_response

3.14

patient.radiations.radiation.radiation_type

patient.radiations.radiation.regimen_indication

3.11

patient.radiations.radiation.units

patient.residual_tumor

patient.stage_event.pathologic_stage

patient.stage_event.system_version

2.81

3.78

patient.stage_event.tnm_categories.pathologic_categories.pathologic_m

patient.stage_event.tnm_categories.pathologic_categories.pathologic_n

patient.stage_event.tnm_categories.pathologic_categories.pathologic_t

patient.synchronous_colon_cancer_present

3.21

3.47

patient.tissue_prospective_collection_indicator

7.16

5.71

4.95

6.44

6.22

4.97

3.96

4.52

4.06

5.58

patient.tissue_retrospective_collection_indicator

7.16

5.71

4.95

6.44

6.22

4.97

3.96

4.52

4.06

5.58

patient.tissue_source_site

1.68

2.00

1.71

patient.venous_invasion

patient.vital_status

3.25

3.97

patient.microsatellite_instability_test_results.microsatellite_instability_test_result.mononucleotide_and_dinucleotide_ma

patient.bcr_canonical_check.bcr_patient_canonical_reasons.bcr_canonical_reason

patient.bcr_canonical_check.bcr_patient_canonical_reasons.bcr_canonical_reason.2

patient.biospecimen_cqcf.consent_or_death_status

patient.biospecimen_cqcf.country

2.16

2.56

2.72

2.55

2.53

2.48

2.93

patient.biospecimen_cqcf.digital_image_submitted

patient.biospecimen_cqcf.frozen_specimen_anatomic_site

patient.biospecimen_cqcf.histological_type

3.20

patient.biospecimen_cqcf.history_of_neoadjuvant_treatment

patient.biospecimen_cqcf.normal_controls.normal_control.2.normal_tissue.normal_tissue_anatomic_site

5.03

4.37

5.87

5.39

5.46

3.77

6.09

6.50

3.35

6.49

3.70

patient.biospecimen_cqcf.normal_controls.normal_control.3.normal_tissue.normal_tissue_anatomic_site

4.24

patient.biospecimen_cqcf.normal_controls.normal_control.extracted_dna.ncedna_dna_qm

3.47

3.03

3.71

3.55

2.98

3.54

3.35

3.72

2.99

patient.biospecimen_cqcf.normal_controls.normal_control.normal_tissue.normal_tissue_anatomic_site

patient.biospecimen_cqcf.other_vessel_used

patient.biospecimen_cqcf.path_confirm_report_attached

patient.biospecimen_cqcf.path_confirm_tumor_necrosis_metrics

patient.biospecimen_cqcf.path_confirm_tumor_nuclei_metrics

patient.biospecimen_cqcf.site_of_disease_description

patient.biospecimen_cqcf.submitted_for_lce

3.72

patient.biospecimen_cqcf.top_slide_submitted

patient.biospecimen_cqcf.vessel_used

patient.samples.sample.2.pathology_report_file_name

patient.samples.sample.2.sample_type

patient.samples.sample.3.sample_type

patient.samples.sample.oct_embedded

patient.samples.sample.pathology_report_file_name

MSI_status

methylation_subtype

anatomic_organ_subdivision

tumor_site

distant_metastasis_pathologic_spread

history_of_colon_polyps

icd_o_3_site

lymphatic_invasion_present

lymphnode_pathologic_spread

person_neoplasm_cancer_status

primary_tumor_pathologic_spread

prior_diagnosis

residual_tumor.y

synchronous_colon_cancer_present

tumor_stage

vascular_invasion_present

vital_status.y

Calculate Wilcoxon Test for binomial Character Variables

```
#Remove factors without enough y-observations
for (i in seq_len(ncol(binary_factor))) {
  x <- summary(binary_factor[,i])

  if (all(x > 1)) {
    print(x)
  } else {
    print(paste("index", i, "does not have enough observations"))
  }
}
```

```
## female   male
##     56     84
## consented  deceased
##       130        10
## [1] "index 3 does not have enough observations"
## female   male
##     56     84
##  no yes
## 127  13
##        canonical canonical - plus
##              114               26
```

```
##  no yes
## 138   2
##   a   b
## 137   3
## [1] "index 9 does not have enough observations"
## [1] "index 10 does not have enough observations"
## [1] "index 11 does not have enough observations"
## picogreen   uv spec
##        32       108
## bottom    top
##     79     61
## [1] "index 14 does not have enough observations"
```

```
bin_fac <- binary_factor[,-c(3, 9, 10, 11, 14)]

wilcox_test_res <- as.data.frame(matrix(nrow = ncol(bin_fac),
                                ncol = ncol(sampleScore_sub)))

rownames(wilcox_test_res) <- colnames(bin_fac)
colnames(wilcox_test_res) <- colnames(sampleScore_sub)

wtest_coad_wvalue <- wilcox_test_res
wtest_coad_pvalue <- wilcox_test_res

for (i in seq_len(ncol(sampleScore_sub))) {
  for (j in seq_len(ncol(bin_fac))) {
    ## wilcoxon test
    wilcox_test <- wilcox.test(sampleScore_sub[, i] ~ bin_fac[,j], alternative="two.sided")

      ## W value
      wval <- wilcox_test$statistic
      wtest_coad_wvalue[j, i] <- wval

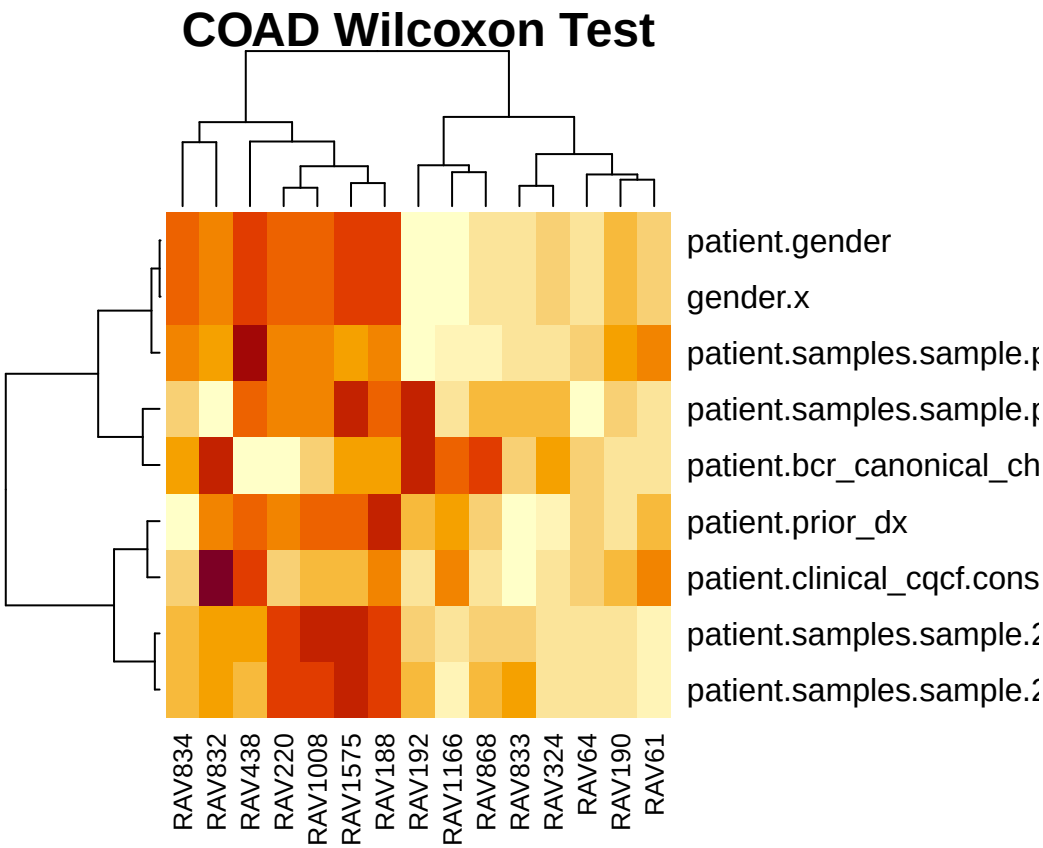
    ## p-value
      pval <- wilcox_test$p.value
      wtest_coad_pvalue[j, i] <- pval

  }
}

# sampleScore_sub[,1]
# bin_fac[,2]

batch_char_ind <- grep('analyte|portion|procurement|aliquot|uuid|barcode',
                  rownames(wtest_coad_wvalue))
wtest_coad_wvalue_2 <- wtest_coad_wvalue[-batch_char_ind,]
```

```
heatmap(as.matrix(wtest_coad_wvalue), main = 'COAD Wilcoxon Test')
```




```
#Only view w-values with significant p-value (<0.05)
wtest_coad_sig_wvalue <- wilcox_test_res


 for (i in seq_len(ncol(sampleScore_sub))) {
   for (j in seq_len(ncol(bin_fac))) {
     if (wilcox.test(sampleScore_sub[, i] ~ bin_fac[,j], alternative="two.sided")$p.val < 0.1) {
       wtest_coad_sig_wvalue[j, i] <- wilcox.test(sampleScore_sub[, i] ~ bin_fac[,j], alternative="two.
     }
   }
 }

# batch_char_ind <- grep('analyte|portion|procurement|aliquot|uuid|barcode',
#                        rownames(wtest_coad_sig_wvalue))
# wtest_coad_sig_wvalue <- wtest_coad_sig_wvalue[-batch_char_ind,]
```

```
options(ztable.type='html')
ztable(wtest_coad_sig_wvalue) %>%
  makeHeatmap(palette = 'Blues') %>%
  print(caption='W-test with p-values < 0.05')
```

W-test with p-values < 0.05

gender.x

NA

NA

NA

NA

NA

NA

patient.clinical_cqcf.consent_or_death_status

NA

NA

NA

NA

NA

NA

patient.gender

NA

NA

NA

NA

NA

NA

patient.prior_dx

NA

NA

NA

NA

NA

NA

526.00

519.00

patient.bcr_canonical_check.bcr_patient_canonical_status

NA

NA

NA

NA

NA

NA

patient.samples.sample.2.is_ffpe

NA

NA

NA

NA

NA

NA

patient.samples.sample.2.vial_number

NA

331.00

NA

NA

72.00

NA

NA

87.00

NA

81.00

90.00

patient.samples.sample.portions.portion.analytes.analyte.spectrophotometer_method

NA

NA

NA

NA

NA

NA

patient.samples.sample.portions.portion.slides.slide.section_location

NA

NA

NA

NA

NA

NA

2854.00

1952.00

```r
# heatmap(as.matrix(wtest_coad_sig_wvalue), main = 'COAD Wilcoxon Test')
```

```r
## Function to Display Categorical Attributes' Score Plots

graph_categorical <- function(a, b, c, d = 'Legend') {
  sampleScore1 <- paste0('RAV', a)
  sampleScore2 <- paste0('RAV', b)

  plot_data <- sampleScore_sub[, c(sampleScore1, sampleScore2)]
  colnames(plot_data)[1] <- "sampleScore1"
  colnames(plot_data)[2] <- "sampleScore2"

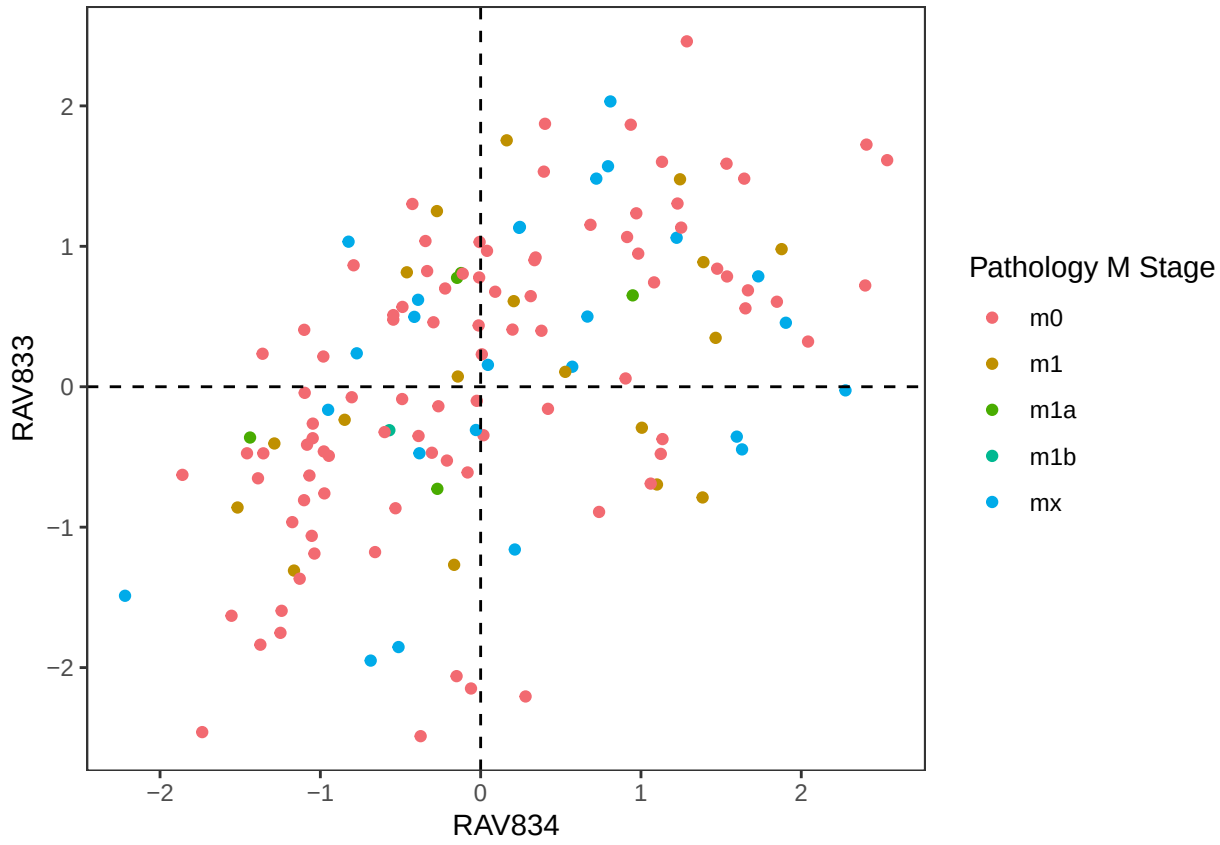
  plot_data <- data.frame(plot_data, new_factorTb)
  plot_data_2 <- plot_data[which(!is.na(factor(plot_data[[c]]))),]
  #plot_data_2 <- plot_data[-which(is.na(factor(plot_data[[c]]))),]

  colors <- gg_color_hue(length(unique(plot_data_2[[c]])))
  colors.toplot <- c(colors)

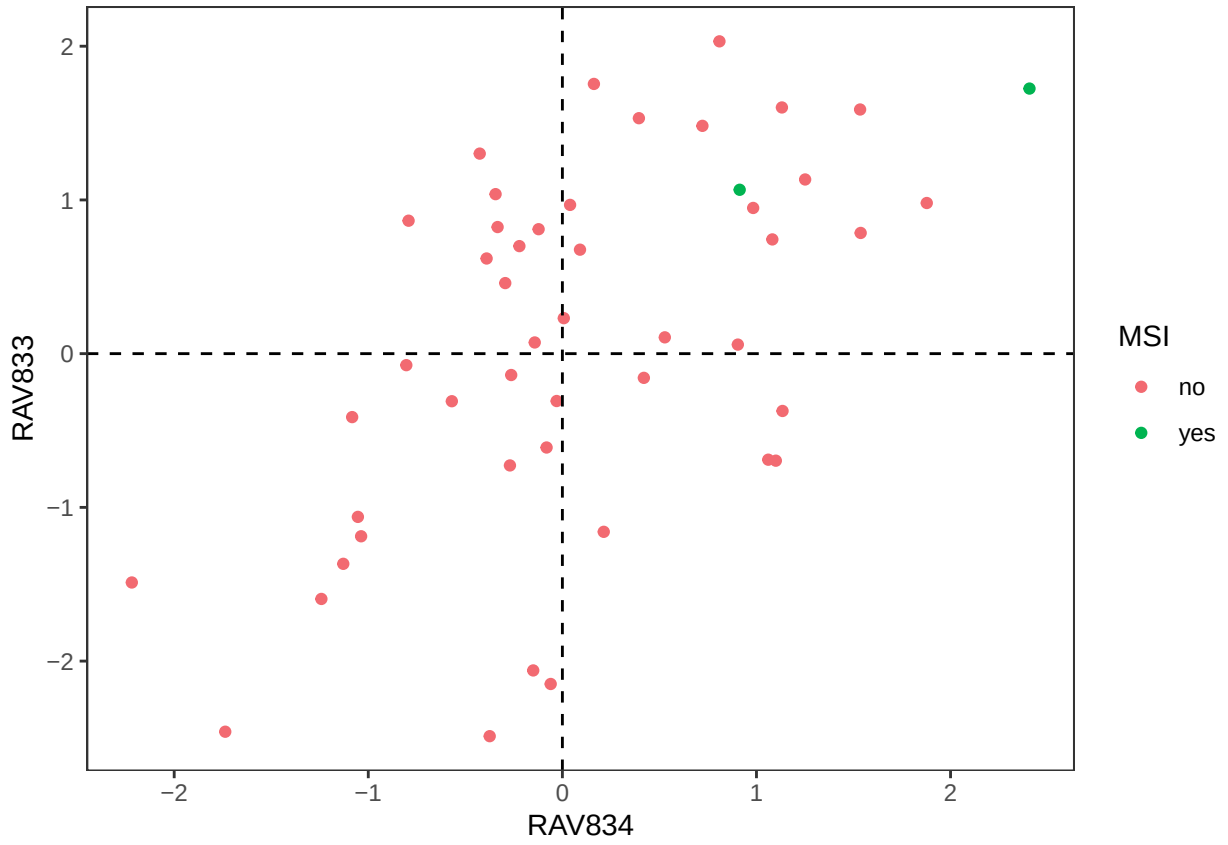
  pA <- ggplot(plot_data_2,
               aes(x = sampleScore1, y = sampleScore2, color = plot_data_2[[c]]
                   )) +
        geom_point() +
        #ggtitle('Score Plot for Gender') +
        scale_color_manual(values = colors.toplot, name = d) +
        theme_bw() +
        theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank()) +
        geom_hline(yintercept = 0, linetype = 'dashed') +
        geom_vline(xintercept = 0, linetype = 'dashed') +
        xlab(sampleScore1) + ylab(sampleScore2)

print(pA)
}
```

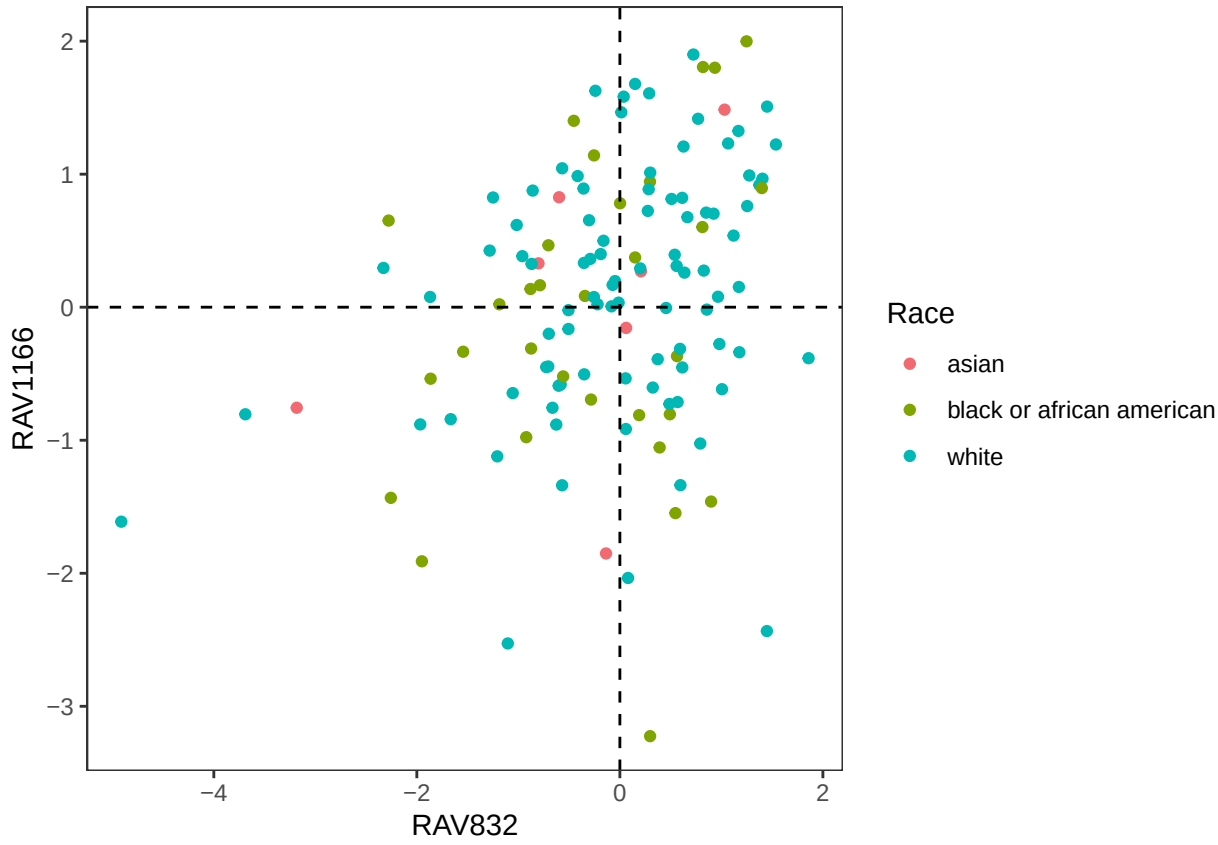
```r
graph_categorical(834, 833, 'pathology_M_stage', 'Pathology M Stage')
```



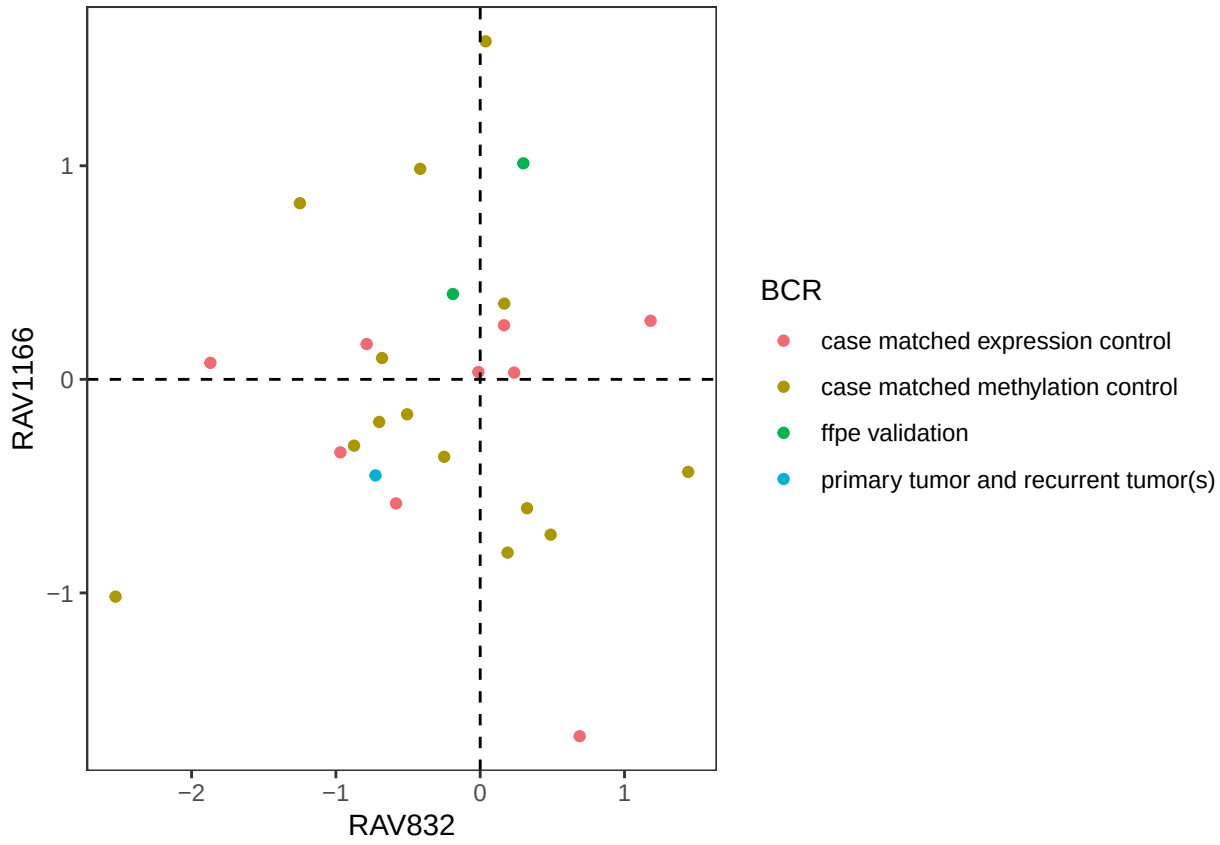

```
graph_categorical(834, 833, 'patient.microsatellite_instability', 'MSI')
```




```
graph_categorical(832, 1166, 'race', 'Race')
```




```
graph_categorical(832, 1166, 'patient.bcr_canonical_check.bcr_patient_canonical_reasons.bcr_canonical_r
```


RAV1166
RAV832
1
0
−1
−2
−1
0
1
BCR
case matched expression control
case matched methylation control
ffpe validation
primary tumor and recurrent tumor(s)